

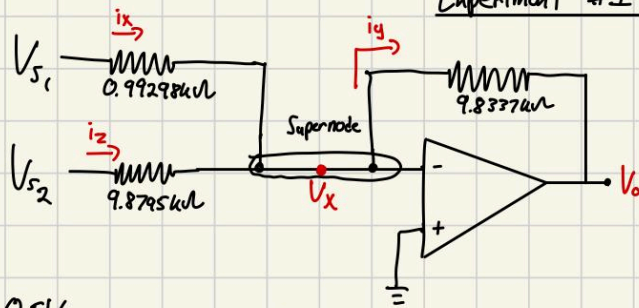
Frank Tamburro

Lab #5

Experiment 1: Summing Amplifier

Experiment #1		Measured	Simulated	Calculated
Resistor Valeus	V_o	-3.9473V	-3.955V	-3.956V
$R_1=0.99298\text{KOHM}$	V_x	1.159mV	39.368microV	0V
$R_2=9.8337\text{KOHM}$	i_x	0.5086mA	503.495microA	0.503mA
$R_3=9.8795\text{KOHM}$	i_y	0.4070mA	402.192microA	0.402mA
	i_z	-0.1061mA	-101.224microA	-0.101mA

Experiment #2



$$V_{s1} = 0.5V$$

$$V_{s2} = -1V$$

$$V_p = V_n = 0V$$

$$V_n = V_x = 0V$$

$$KCL: V_x$$

$$i_x - i_y + i_z = 0$$

$$i_x + i_z = i_y$$

$$\frac{0.5V - 0V}{0.99298k\Omega} + \frac{-1V - 0V}{9.8795k\Omega} = \frac{0 - V_o}{9.8337k\Omega}$$

$$i_x = 0.503mA$$

$$i_y = 0.402mA$$

$$i_z = -0.101mA$$

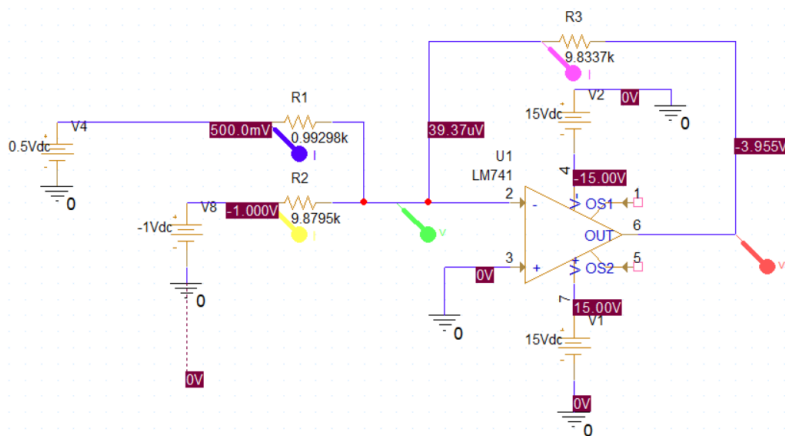
$$\frac{0.5V}{0.99298k\Omega} - \frac{1V}{9.8795k\Omega} = \frac{-V_o}{9.8337k\Omega}$$

$$0.503mA - 0.101mA = \frac{-V_o}{9.8337k\Omega}$$

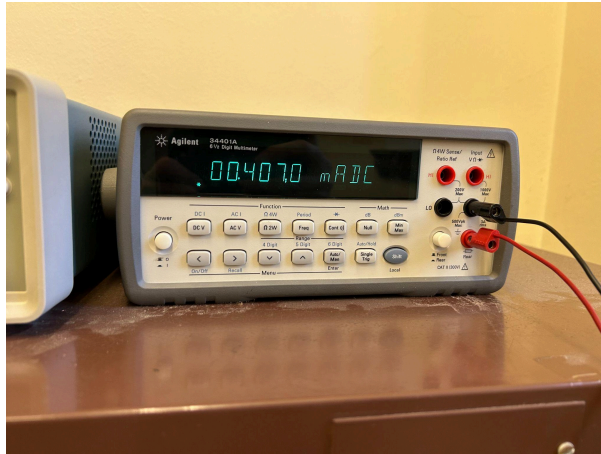
$$(0.402mA = \frac{-V_o}{9.8337k\Omega}) \cdot 9.8337k\Omega$$

$$3.956 = -V_o$$

$$V_o = -3.956V$$

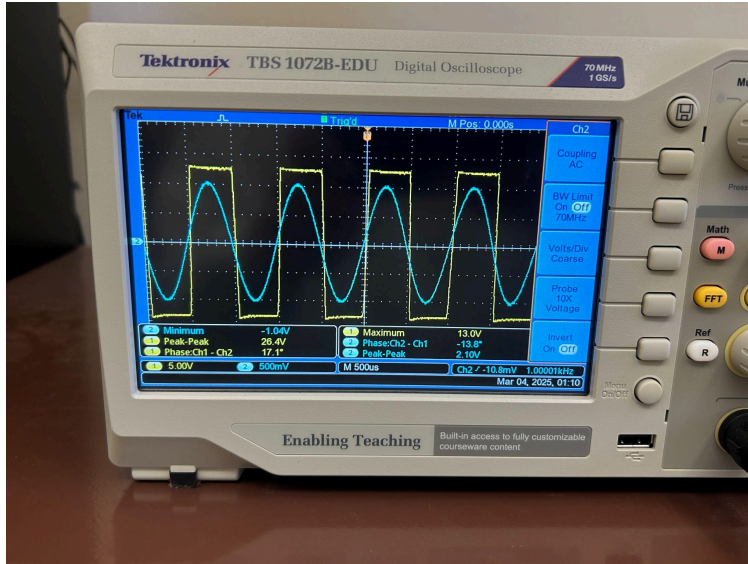






Conclusion: I believe that this experiment was successful because all of my measured, simulated, and calculated values are all very close to each other. There is not one value that is significantly different from each other.

Experiment 2: Comparator



Experiment #2

$R1 = 0.99298 \text{ KOHM}$

$R2 = 0.98949 \text{ KOHM}$

Conclusion: The output of this experiment is the yellow square wave. With the sinusoidal input (blue wave) the operation amplifier changes the input sine wave into a square wave because it's an open loop configuration (comparator). This causes the outputs to jump from -15V to 15V and to have flat parts at the voltages for a period of time.

Experiment 3: Unity Gain Amplifier and Op-Amp Slew Rate Measurement

Experiment #3				
RS=0.99298KOHM				
RL=0.98949KOHM				
		Measured	Simulated	Calculated
Circuit A	Ia	1.0270mA	1.0088mA	1.00mA
	Va	1.01544V	0.9982V	1.00V
Circuit B	Ia	0A	80.600nA	0A
	Va	0.9411V	1.999V	1.00V
	Slew Rate	2.042M(V/S)		

Experiment #3

Circuit (a)

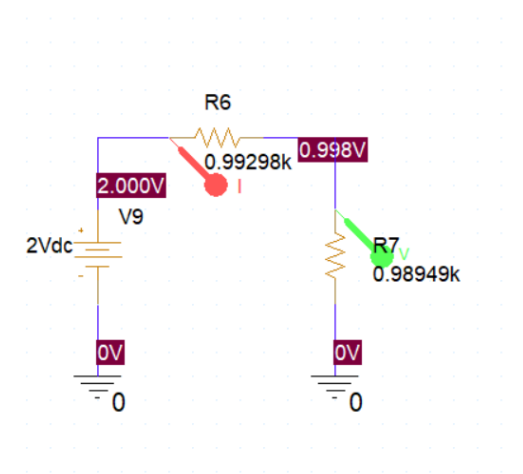
$R_S = 0.99298k\Omega$
 $R_L = 0.98949k\Omega$
 $i_a = \frac{V}{R} = \frac{2}{0.99298k\Omega + 0.98949k\Omega}$
 $i_a = 1.00mA$
 $V_a = V_f \left(\frac{0.98949}{0.99298 + 0.98949} \right)$
 $V_a = 2V \left(\frac{0.98949}{0.99298 + 0.98949} \right)$
 $V_a = 1.00V$

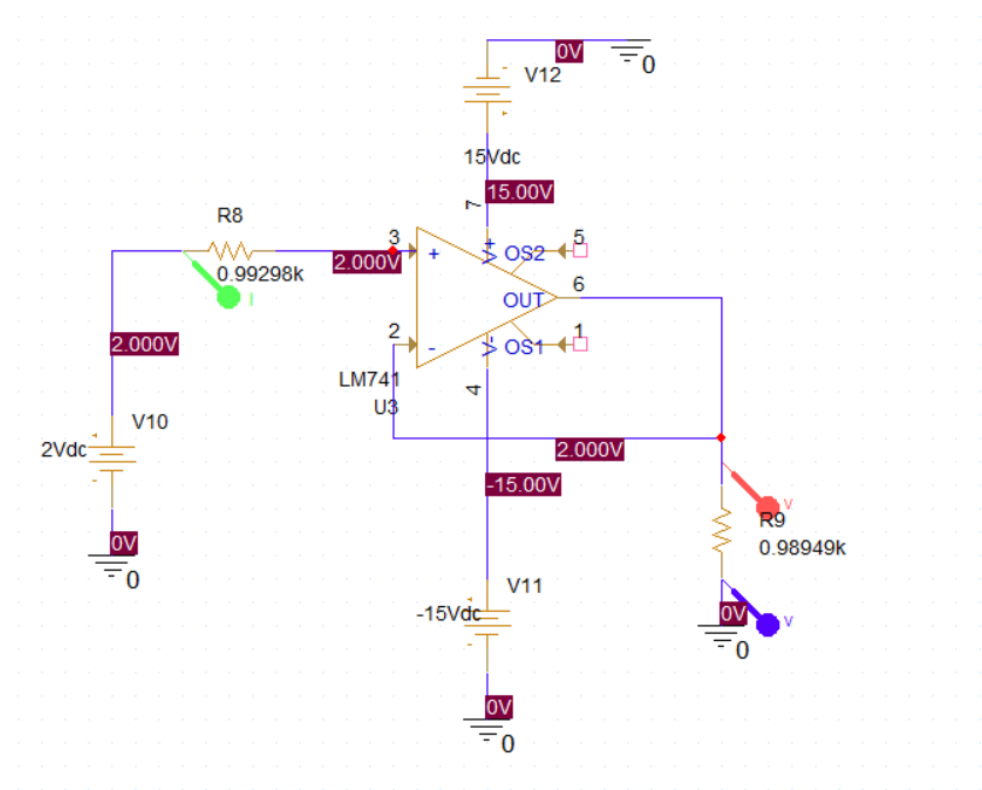
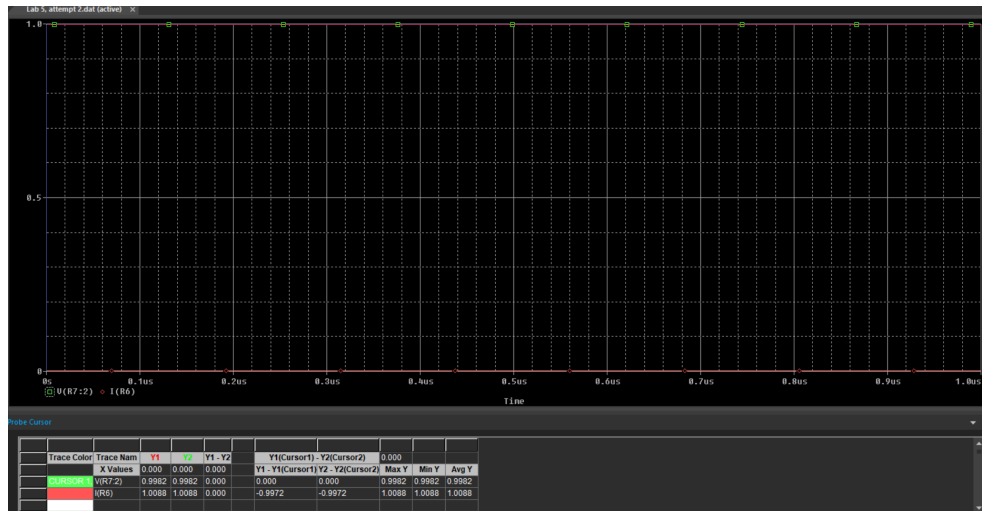
Circuit (b)

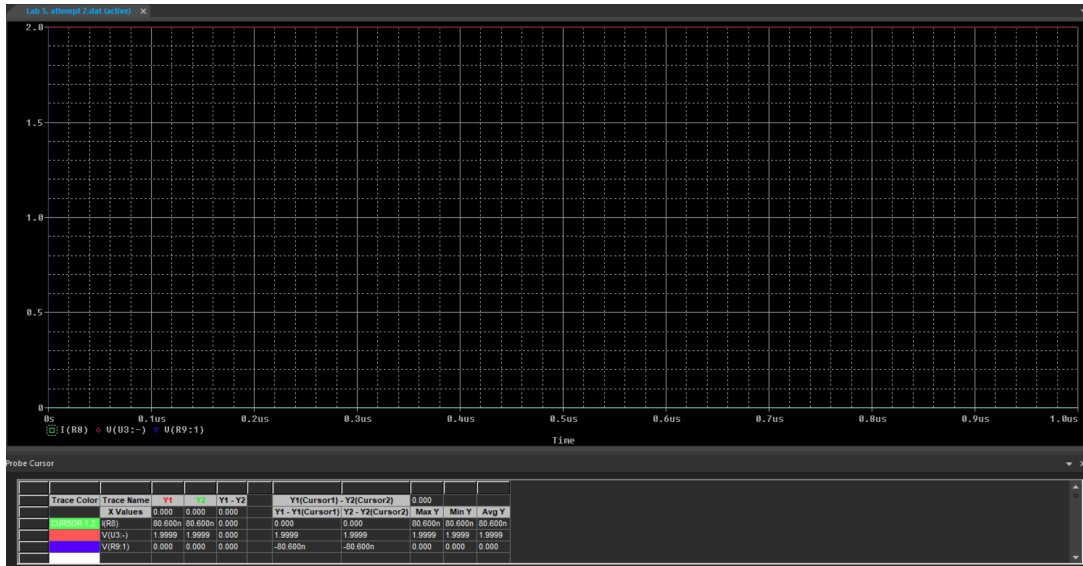
$R_S = 0.99298k\Omega$
 $R_L = 0.98949k\Omega$
 $i_p = i_n = 0A$
 $i_a = 0A$
 $V_p = V_n = V_a$
 $\left(\frac{2 - V_a}{0.99298k\Omega} = \frac{V_a - 0V}{0.98949k\Omega} \right) 0.98949k\Omega$
 $0.996(2 - V_a) = V_a$
 $1.993 - 0.996V_a = V_a$
 $\frac{1.993}{1.996} = \frac{1.996V_a}{1.996}$
 $V_a = 1.00V$

Slew rate: $2 \pi f V$

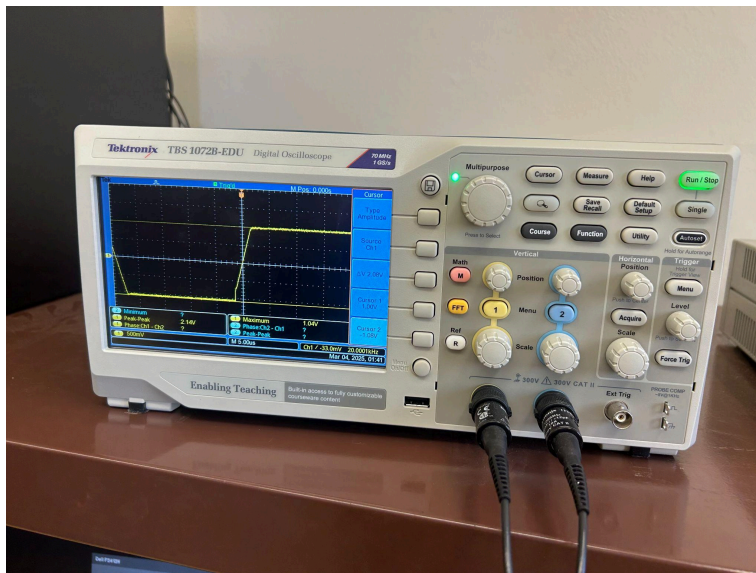
$\Delta t = 3.20 \mu s$
 $\Delta V = 2.08V$
 $T = 2 \cdot \Delta t$
 $T = 6.40 \mu s$
 $f = \frac{1}{T} = \frac{1}{6.40 \mu s} = 156.25 kHz$
 $Slew\ rate = 2 \pi (156.25 kHz)(2.08V)$
 $Slew\ rate = 2042035.225 \frac{V}{s}$
 $Slew\ rate = 2.042 M \frac{V}{s}$











Conclusion: In circuit A all of the values are very similar and result in a successful experiment. In circuit B all of the values are very similar to each other except for one in circuit B. The V_a 's simulated value is double the calculated and measured value.

Experiment 4: Voltage to Current Converter

Experiment #4									
Part 2	Vs(Volts)	Measured(i1)	Simulated(i1)	Calculated (i1)	Part 2	Vs(Volts)	Measured(i1)	Simulated(i1)	Calculated (i1)
R1=3.2618KOHM	-6V	-1.8755mA	-1.8455mA	-1.845mA	R1=9.8806KOHM	-6V	-0.9828mA	-1.1282mA	-1.845mA
R2=3.2509KOHM	-5V	-1.5584mA	-1.5379mA	-1.538mA	R2=3.2509KOHM	-5V	-0.9831mA	-1.1282mA	-1.538mA
	-4V	-1.2314mA	-1.2303mA	-1.2304mA		-4V	-0.9829mA	-1.1282mA	-1.2304mA
	-3V	-0.9379mA	-0.9167mA	-0.922mA		-3V	-0.9375mA	-0.92268mA	-0.922mA
	-2V	-0.6203mA	-0.6151mA	-0.615mA		-2V	-0.6251mA	-0.60918mA	-0.615mA
	-1V	-0.3133mA	-0.3075mA	-0.3076mA		-1V	-0.3169mA	-0.30751mA	-0.3076mA
	0V	-0.0018mA	-0.0858mA	0A		0V	0.0005mA	0.085684mA	0A
	1V	0.0313mA	0.3076mA	0.3076mA		1V	0.3106mA	0.30767mA	0.3076mA
	2V	0.6266mA	0.6152mA	0.6152mA		2V	0.6215mA	0.61527mA	0.6152mA
	3V	0.9339mA	0.9228mA	0.922mA		3V	0.9399mA	0.92286mA	0.922mA
	4V	1.2459mA	1.2305mA	1.2304mA		4V	1.1096mA	1.1282mA	1.2304mA
	5V	1.5566mA	1.5381mA	1.538mA		5V	1.1096mA	1.1282mA	1.538mA
	6V	1.8694mA	1.8455mA	1.845mA		6V	1.1096mA	1.1282mA	1.845mA

Experiment 4

Circuit (a)

V_s ranges by -6V to 6V by 1V increments
 $V_p = V_n = V_s$
 $I_n = I_p = 0A$
 $I_1 = I_L$

$$\frac{V_{out} - V_n}{3.2618k\Omega} = \frac{V_n - 0}{3.2509k\Omega}$$

$$\frac{V_{out} - V_n}{3.2618k\Omega} = \frac{V_n}{3.2509k\Omega}$$

$A_1 = 3.2618k\Omega$
 $A_2 = 3.2509k\Omega$

Circuit (b)

V_s ranges by -6V to 6V by 1V increments
 $V_p = V_n = V_s$
 $I_n = I_p = 0A$
 $I_1 = I_L$

$$\frac{V_{out} - V_n}{9.8806k\Omega} = \frac{V_n - 0}{3.2509k\Omega}$$

$$\frac{V_{out} - V_n}{9.8806k\Omega} = \frac{V_n}{3.2509k\Omega}$$

$A_1 = 9.8806k\Omega$
 $A_2 = 3.2509k\Omega$

$$a) \quad \frac{U_{out} - U_{in}}{2.2618 \mu A} = \frac{U_{in}}{2.2509 \mu A}$$

$$-6V) \quad \frac{U_{out} + 6}{2.2618 \mu A} = \frac{-6}{2.2509 \mu A}$$

$$I_1 = -1.895 \text{ mA}$$

$$6V) \quad \frac{U_{out} - 6}{2.2618 \mu A} = \frac{6}{2.2509 \mu A}$$

$$I_1 = 1.845 \text{ mA}$$

$$-9V) \quad \frac{U_{out} + 9}{2.2618 \mu A} = \frac{-9}{2.2509 \mu A}$$

$$I_1 = -1.538 \text{ mA}$$

$$-4V) \quad \frac{U_{out} + 4}{2.2618 \mu A} = \frac{-4}{2.2509 \mu A}$$

$$I_1 = -1.2304 \text{ mA}$$

$$-3V) \quad \frac{U_{out} + 3}{2.2618 \mu A} = \frac{-3}{2.2509 \mu A}$$

$$I_1 = -0.922 \text{ mA}$$

$$-2V) \quad \frac{U_{out} + 2}{2.2618 \mu A} = \frac{-2}{2.2509 \mu A}$$

$$I_1 = 0.615 \text{ mA}$$

$$-1V) \quad \frac{U_{out} + 1}{2.2618 \mu A} = \frac{-1}{2.2509 \mu A}$$

$$I_1 = -0.3076 \text{ mA}$$

$$0V) \quad \frac{U_{out} - 0}{2.2618 \mu A} = \frac{0}{2.2509 \mu A}$$

$$I_1 = 0 \text{ A}$$

$$1V) \quad \frac{U_{out} - 1}{2.2618 \mu A} = \frac{1}{2.2509 \mu A}$$

$$I_1 = 0.3076 \text{ mA}$$

$$2V) \quad \frac{U_{out} - 2}{2.2618 \mu A} = \frac{2}{2.2509 \mu A}$$

$$I_1 = 0.615 \text{ mA}$$

$$3V) \quad \frac{U_{out} - 3}{2.2618 \mu A} = \frac{3}{2.2509 \mu A}$$

$$I_1 = -0.922 \text{ mA}$$

$$4V) \quad \frac{U_{out} - 4}{2.2618 \mu A} = \frac{4}{2.2509 \mu A}$$

$$I_1 = 1.2304 \text{ mA}$$

$$5V) \quad \frac{U_{out} - 5}{2.2618 \mu A} = \frac{5}{2.2509 \mu A}$$

$$I_1 = 1.538 \text{ mA}$$

$$\frac{V_{out} - V_n}{9.8806 \mu A} = \frac{V_n}{3.2509 \mu A}$$

$$6V) \frac{V_{out} - 6}{9.8806 \mu A} = \frac{6}{3.2509 \mu A}$$

$$I_1 = 1.845 mA$$

$$-6V) \frac{V_{out} + 6V}{9.8806 \mu A} = \frac{-6}{3.2509 \mu A}$$

$$I_1 = -1.845 mA$$

$$-5V) \frac{V_{out} + 5V}{9.8806 \mu A} = \frac{-5}{3.2509 \mu A}$$

$$I_1 = -1.538 mA$$

$$-4V) \frac{V_{out} + 4V}{9.8806 \mu A} = \frac{-4V}{3.2509 \mu A}$$

$$I_1 = -1.2304$$

$$-3V) \frac{V_{out} + 3}{9.8806 \mu A} = \frac{-3}{3.2509 \mu A}$$

$$I_1 = -0.922 mA$$

$$2V) \frac{V_{out} + 2}{9.8806 \mu A} = \frac{-2}{3.2509 \mu A}$$

$$I_1 = 0.615 mA$$

$$-1V) \frac{V_{out} + 1}{9.8806 \mu A} = \frac{-1}{3.2509 \mu A}$$

$$I_1 = -0.3076 mA$$

$$0V) \frac{V_{out} - 0}{9.8806 \mu A} = \frac{0}{3.2509 \mu A}$$

$$I_1 = 0 A$$

$$1V) \frac{V_{out} - 1}{9.8806 \mu A} = \frac{1}{3.2509 \mu A}$$

$$I_1 = 0.3076 mA$$

$$2V) \frac{V_{out} - 2}{9.8806 \mu A} = \frac{2}{3.2509 \mu A}$$

$$I_1 = 0.615 mA$$

$$3V) \frac{V_{out} - 3}{9.8806 \mu A} = \frac{3}{3.2509 \mu A}$$

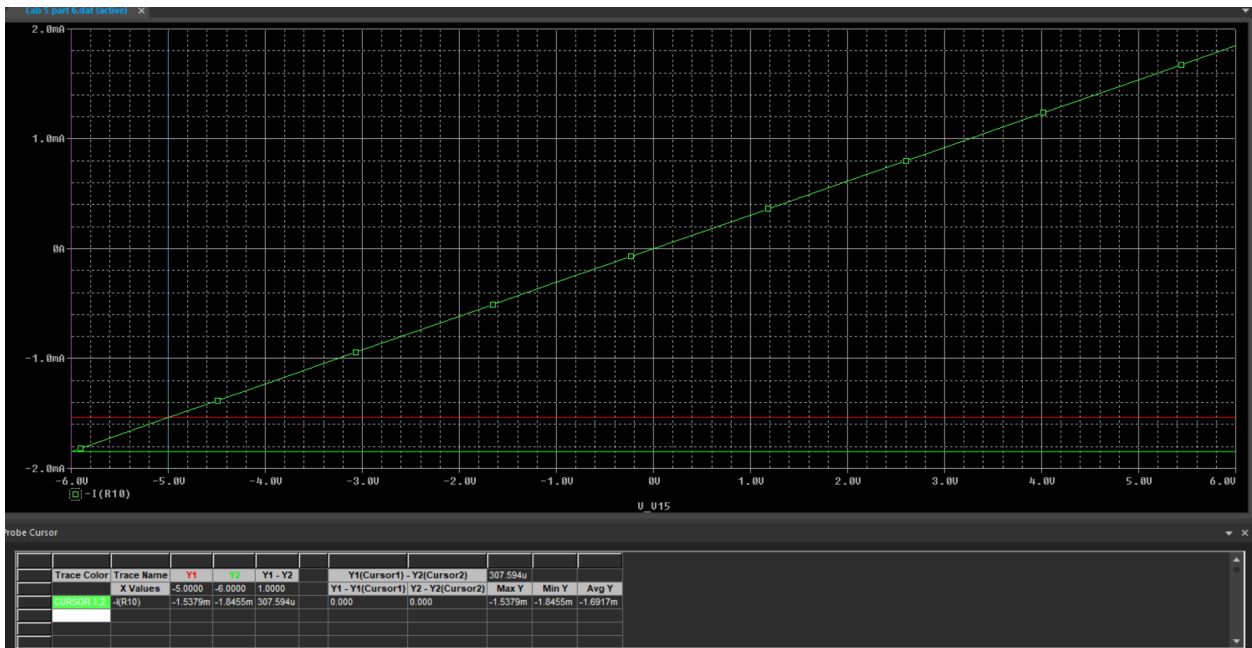
$$I_1 = -0.922 mA$$

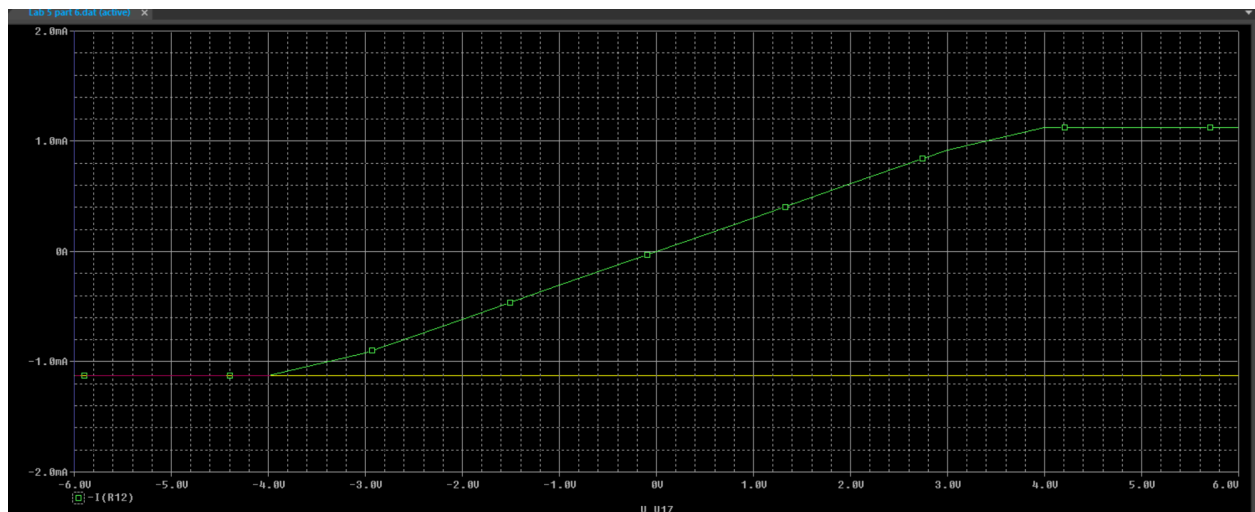
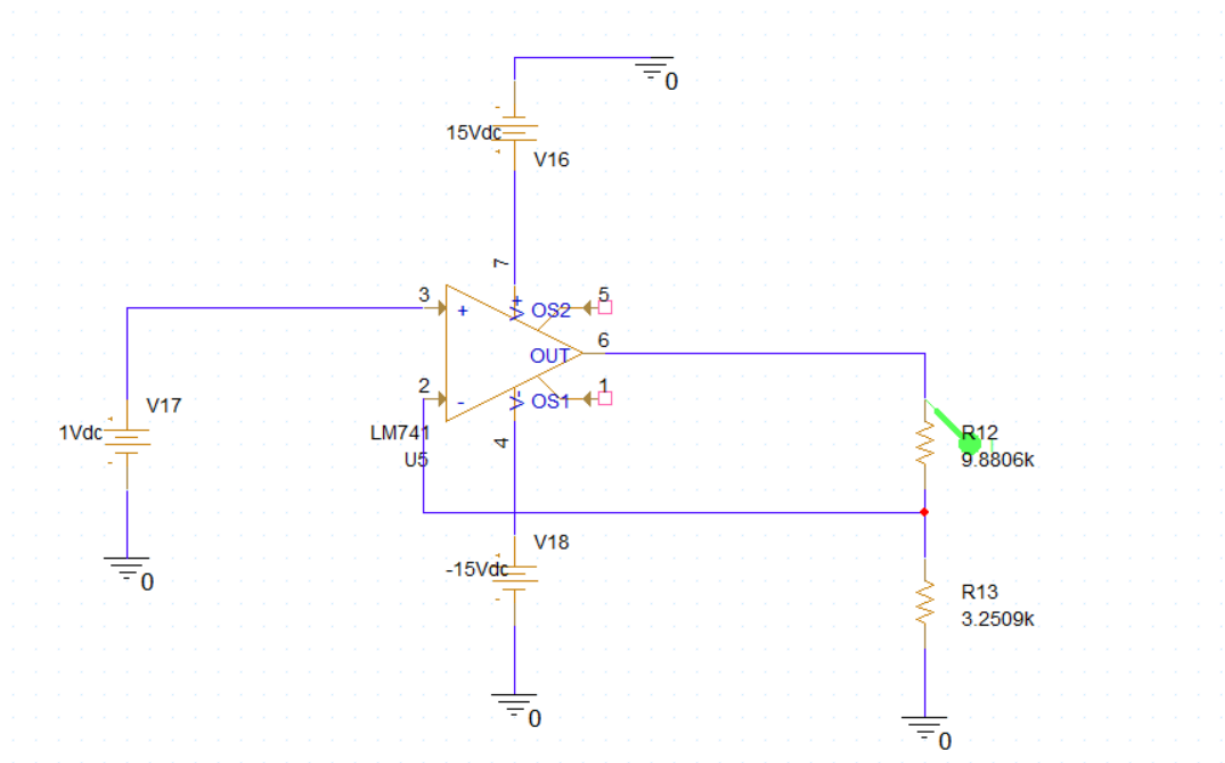
$$4V) \frac{V_{out} - 4}{9.8806 \mu A} = \frac{4}{3.2509 \mu A}$$

$$I_1 = 1.2304 mA$$

$$5V) \frac{V_{out} - 5}{9.8806 \mu A} = \frac{5}{3.2509 \mu A}$$

$$I_1 = 1.538 mA$$





Conclusion: In part 1 of the experiment all values were the same and all acted on a linear trend line like it shows in the screenshot from Pspice. However in part 2, the trend line had flat parts at the beginning and end. This is accurate for the measured and simulated values, but in the calculations the numbers went un linearly instead of being the same for the first and last 3 voltages.